THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 8: **Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An echo is heard after 0.4 s at 20 degrees Celsius (v = 344 m/s). The distance of the reflector is: 1

(a) 344 m **(b)** 137.6 m **(c)** 34.4 m **(d)** 68.8 m

2. A mosquito buzz has a higher pitch than the roar of a lion because it has: 1

(a) lower frequency **(b)** higher amplitude
(c) higher frequency **(d)** greater wavelength

3. A source produces a sound of 25000 Hz. This is: 1

(a) ultrasonic **(b)** infrasonic **(c)** noise **(d)** audible

4. Ultrasonic waves are used for cleaning because: 1

(a) they are loud
(b) they have long wavelengths
(c) their high frequency vibrations dislodge dirt
(d) they heat the object

5. A pulse takes 2.4 s to go to the sea bed and back at v = 1500 m/s. The depth is: 1

(a) 1800 m **(b)** 625 m **(c)** 3600 m **(d)** 900 m

6. When a sound wave passes from air into water, which quantity remains the same? 1

(a) speed **(b)** amplitude **(c)** frequency **(d)** wavelength

7. Dogs can hear up to 45 kHz. A whistle of 30 kHz is: 1

(a) audible to humans **(b)** infrasonic
(c) inaudible to both **(d)** audible to dogs but not to humans

8. The SI unit of the time period of a wave is: 1

(a) decibel **(b)** metre **(c)** hertz **(d)** second

9. Sound travels from a source to a listener 1.7 km away in: 1

(a) 0.005 s **(b)** 5 s **(c)** 50 s **(d)** 0.5 s

10. The wavelength of a 1000 Hz sound in water (v = 1500 m/s) is: 1

(a) 0.67 m **(b)** 1500 m **(c)** 1.5 m **(d)** 15 m

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do whales communicate over long distances in the sea? 2

12. Why does sound travel faster in warm air than in cold air? 2

13. Why is a sound of 15 Hz not audible while 15 kHz is? 2

14. A drum is struck harder. Which characteristic of the sound changes and how? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the reflection of sound with the laws and the uses of curved reflectors (megaphone, ear trumpet, sound boards). 3

16. The frequency of a vibrating guitar string is 440 Hz. Find the number of vibrations in 1 minute and the time for one vibration. 3

17. Calculate the wavelength of the lowest and the highest audible notes in air (v = 340 m/s). 3

OR

Explain how a doctor uses ultrasound for imaging organs and why it is safer than X-rays. 3

18. A sound is received 0.5 s after it is emitted in water and the same in steel for the same path. Compare the distances using v = 1500 m/s in water and 5000 m/s in steel. 3

OR

State the changes in a wave's amplitude, frequency and wavelength when it travels from air to water. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A sound of 1700 Hz is produced in air. Find its wavelength and the number of waves in 34 m and the time for sound to travel 34 m. Then repeat for a medium where v = 1700 m/s. 5

OR

Compare light and sound waves in a table of five points and give two phenomena that both show. 5

20. Discuss the applications of reflection of sound: megaphones, stethoscopes, sound boards and the design of concert halls, with the physics of each. 5

OR

Explain how the wavelength, frequency and speed change when a 440 Hz sound goes from air (340 m/s) into water (1500 m/s) and into steel (5000 m/s). Calculate the wavelengths. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND